Fish Schooling and Predator Evasion Under The Cell-DEVS Formalism

SYSC4906G

Zachary Dredge – 101197514

April 24th, 2026

Introduction

Fish schooling is one of the most widely studied examples of collective behaviour in nature. With each fish following simple local rules based on their surroundings, they manage to produce globally coordinated movement patterns that serve as a critical survival mechanism. Perhaps the most important of these patterns is predator evasion: where when attacked the school moves in a synchronized manner that confuses predators and protects the individuals of the school.

This project presents an asymmetric Cell-DEVS model of fish school formation and predator evasion, implemented in the Cadmium simulation framework. The model extends prior work by Etemad, which simulated the formation of schools using symmetric Cell-DEVS and was implemented in CD++ [1]. As suggested in the original work, the new predator evasion behaviour logic will be drawn from those identified of Zheng et al. [2]. Additionally, the model will also incorporate the idea of water currents, which will serve as asymmetric influences on fish and predator movement.

Motivations

Schools of fish reduce individual predation risk through several factors, including coordinated confusion – where a predator facing a rapidly moving group of fish has difficulty singling out and tracking a target [2]. When a predator can isolate a fish from its school, the probability of capture greatly increases. Due to this, the school's collective evasion strategy is critical to the survival of its members.

Prior Cell-DEVS modeling of fish schools as done in the prior assignments this work extends, focused solely on the formation of schools in environments without the presence of a predator. Both works simulate how distributed fish discover their neighbors, form schools, and anchor in place [1]. However, these models do not consider the presence of a predator, which is the very reason behind schooling behaviour. As such, through this work we seek to model not just the formation, but also the predator-prey interaction driving schooling behaviour.

Goals

This project has three primary goals:

1. Convert the existing symmetric Cell-DEVS fish school model into an asymmetric Cell-DEVS system.
2. Introduce a predator to the system and implement three-pattern evasion behaviour (schooling, cooperative escape, and selfish escape) described by Zheng et al. [2].
3. Model environmental water currents as directional asymmetric vicinity factors, thereby demonstrating how currents influence both fish evasion strategies and predator pursuit.

Contributions

The major contributions of this work are as follows.

Firstly, we will present a formal specification of an asymmetric Cell-DEVS model that combines fish schooling, predator pursuit, and environmental current factors. The fish and predator

components will be modeled as separate cell types with distinct neighborhood configurations. They will however share a local computation function as this is simpler to implement their interactions this way.

Secondly, we will adapt the work done by Zheng et al. to a Cell-DEVS formalism, mapping the distance-based behaviour switching rule to a cell-range threshold-based system [2].

Thirdly, we will introduce a vicinity factor to encode water current effects, where the vicinity value on each directed cell connection represents the current's assistance or opposition to movement along that connection.

Finally, we will provide an experimental framework that serves to verify and validate the interactions within the grid space. With this, we will track a metric used to show how many of the simulations result in open-water captures, that being captures that do not occur along the border of the grid space.

Background

Fish Schooling Behaviour

Fish schools are characterized by its members that maintain a common direction, regular spacing, and uniform movement speed [3]. This is achieved through individual organization, rather than having a central leader. That being said, individual fish determine their movement based on their immediate neighbors, and by following three interaction rules; mimicking the direction and speed of nearby fish, collision avoidance, and attraction [4] [5].

In a reference paper for this work, Hubbard et al. modeled fish school formation and migration, and observed that the members maintain constant speed and direction during migration but may remain stationary for longer periods [3]. They also noted that thermal boundaries play a role in the movement of the school, and it is unlikely the boundaries are crossed [3]. For the purposes of this project, we will not consider thermal boundaries and instead make the following assumption: that fish schools will remain anchored in place unless they are exhibiting cooperative escape behaviour, as described below. However, we will take inspiration from this idea, and implement a current-based influence on fish movement.

Predator Evasion in Fish Schools

Zheng et al. present a theoretical model of individual behaviour patterns that generate efficient collective evasion in fish schools [2]. Their model defines three component behaviour patterns (BP) that each fish chooses based on the predator's proximity.

1. **Schooling BP:** When no predator is within visible range, or when a predator is visible but beyond a certain distance, fish continue normal schooling behaviour [2]. This consists of allelomimetic motion (mimicking neighbors) [2].
2. **Cooperative Escape BP:** When the predator is within the cautionary distance but beyond the immediate threat distance, where fish combine school-formation movement with

fleeing from a predator [2]. The critical thing here is that fish prioritize maintaining school cohesion over pure flight, due to the fact that isolated fish are easily captured [2].

3. **Selfish Escape BP:** When the predator is within the immediate threat range, where fish begin to act selfishly and attempt to swerve off the predators line of attack [2].

The switch between the BPs is decided by a distance-based rule. Zheng et al. define five concentric zones around each fish relative to the predator: a capture area, an immediate threat area, a caution area, a patience area, and an invisible area [2]. This is shown in Fig. 1. A key finding of their work is the patience parameter ε_{pat} at a value of 0.4 means the fish do not begin their cooperative escape until the predator is within 40% of their maximum visible range. This patience parameter is used to prevent the premature disruption of school structure.

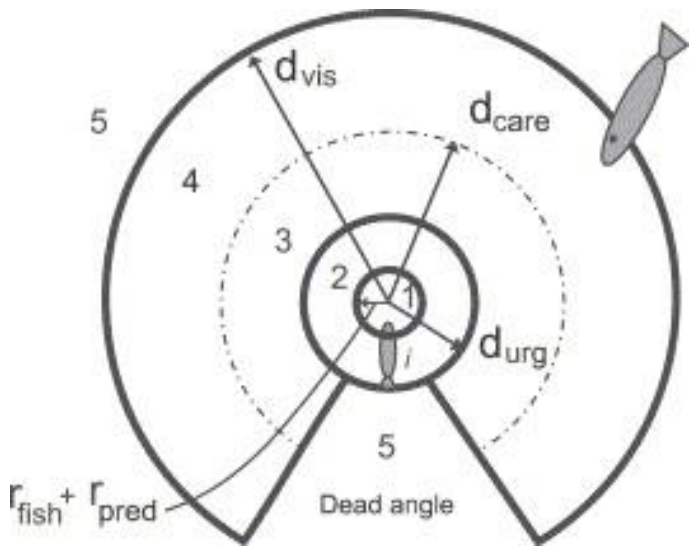

Fig. 1. Fish Vision Range as Defined by Zheng et al. [2]

In Zheng et al. the predator is modeled with a fan-shaped attack zone, where it is able to target a singular fish [2]. The predator is also modeled with limited turning ability, and the tendency to switch target fish when confused [2]. Each of these factors will be considered in our approach.

Cell-DEVS Formalism

Prior Work

Etemad developed the original fish school formation model in CD++, using a three-layer symmetric Cell-DEVS structure on a 10x10 grid [1]. The first layer tracked the fish's current position, the second represented random movement direction, and the third represented fish orientations. In this model, fish moved randomly until detecting a neighbor within their field of view, at which point they began to follow priority-based directional attraction rules. If fish were to form a school, they were then anchored to their assigned cells.

In assignment 2, this initial model was adapted to Cadmium, compressing the three-layer system into a single layer with a state struct containing presence, direction, and orientation fields. The adaptation expanded the fish vision radius to two cells, and implemented anti-chase logic to

prevent cyclic movement patterns. This implementation will serve as the foundation for the present work and is extended with predator evasion and asymmetric Cell-DEVS functionality.

Models Defined

This section details the formal specifications of the asymmetric Cell-DEVS models. The model consists of two cell types – fish cells and predator cells – coupled within an asymmetric Cell-DEVS grid that is 20x20. The grid space also features water current effects that are represented by vicinity factors.

State Set

$S = \{p, d, o, b, predatorDist, predatorDir\}$

Where,

- Presence (p) indicates if the cell is with or without a fish or predator. $p \epsilon \{0, 5, 10\}$, where 0 indicates an empty cell, 5 indicates the presence of a fish, and 10 indicates the presence of a predator.
- Direction (d) indicates the next direction the fish will attempt to move. $d \epsilon \{0, 1,2,3,4\}$, mapping to None, East, North, West, South.
- Orientation (o) indicates the direction the fish was heading in the previous time step. $o \epsilon \{0,1,2,3,4\}$, where 0 indicates no previous movement. The remaining values map the same as d.
- Behaviour (b) indicates the current BP of the fish. $b \epsilon \{0, 1, 2\}$, where $0 = schooling, 1 = cooperative\ escape, 2 = selfish\ escape$.
- Predator Distance $(predatorDist)$ indicates the distance to the nearest predator. This is calculated using Manhattan distance, and a value of 0 indicates no predator is detected.
- Predator Direction $(predatorDir)$ indicates the direction from which a predator is approaching. $predDir \epsilon \{0, 1, 2, 3, 4\}$, mapping to the same values as d.

Vicinity Factor

The vicinity factor V is a double-precision floating-point value that is assigned to each directed neighbor connection. It encodes the water current's effect on movement along that specific connection.

- $V > 0$: Current assists movement in this direction.
- $V = 0$: No current effect
- $V < 0$: Current opposed movement in this direction.

It is important to note that from Cell A to Cell B, and from Cell B to Cell A, the connections will carry opposite-signed vicinity factors. For example, in a region with eastward current, the A to B connection could be $V = +1.0$, while B to A (westward) will be $V = -1.0$.

In the local computation function, fish will use the vicinity factor to bias their movement decisions. This means that when choosing between multiple movement directions, fish will favor directions with higher vicinity factors as they represent “current-assisted” directions. This means that the model will shift from having random fish movement patterns to influenced and calculated movement patterns in the presence of water currents.

Fish Cell Atomic Model

The formal specification for the atomic Cell-DEVS model is defined as follows:

$$< X, Y, S, N, d, \tau, \delta_{int}, \delta_{ext}, \lambda, D >$$

- S: As defined in *State Set*
- $X = \{x | x \epsilon S\}$
- $Y = \{y | y \epsilon S\}$
- $N = S \quad \rightarrow \{\{(dx, dy) | \, |dx| + |dy| \leq 4\}\} \, for \, range - 4 \, Von \, Neumann \, Neighborhoods$
- $D = 1 \, time \, unit$
- $\tau: N \rightarrow S$

Local Computation Function

The fish cell’s local computation function implements the component behaviour pattern with distance-based switching, adapted from Zheng et al. [2]. At each timestep, a fish cell first scans its neighborhood (range 4) for predator presence and computes the Manhattan distance to the nearest predator. The behaviour mode is then selected:

- **Schooling (behaviour = 0):** No predator detected within range 4. The fish follows the existing schooling rules: school detection, anchoring, priority-based attraction (West > East > North > South), orientation-based blind spots, and random movement for isolated fish (in the absence of water currents).

- **Cooperative escape (behaviour = 1):** Predator detected at range 3-4 (caution zone). All cooperating fish compute the same flee direction, that being the opposite direction of the approaching predator. When the primary direction is blocked, the school will fallback to the current-assisted direction (positive vicinity factor).

- **Selfish escape (behaviour = 2):** Predator detected at range 1 or 2 (urgent zone). The fish ignores school cohesion and moves perpendicular to the predator's orientation (movement direction), implementing the swerve behaviour. For example, if the predator is moving East, the fish moves North or South. The school anchor is overridden — the fish vacates even if it was part of a school. Current-assisted directions are strongly preferred in selfish escape.

Alarm Relay Threshold

A fish that directly detects a predator within its vision range updates its $predatorDist$ and $predatorDir$ state fields accordingly. These fields also serve a secondary purpose: they allow predator awareness to propagate through the school to fish that cannot directly see the predator.

At each timestep, a fish that has no direct predators will check two relay sources. First, it will check its own $predatorDist$ value, and if it previously held an alarm then it will increment the distance by one and carry the alarm forward in a self-decaying manner. Second, its checks both fields of each of its neighboring fish and calculates the Manhattan distance from itself to the neighbor, plus the distance to the predator. The current fish will then adopt the minimum value of all its neighbors.

To prevent stale alarms, propagation is limited to a bound of 7. This value allows for a direct detection range of 4, aligned with fish vision, and 3 additional hops to account for the size of the school. A fish that receives only a relayed alarm, and not direct predator detection, will switch to cooperative escape behaviour. The bound on this propagation is best seen in the simulations section, in particular in the *large_school* scenario.

Current Influence on Fish Movement

When a fish has multiple valid movement options, it consults the vicinity factors of the corresponding neighbor connections. The direction with the higher vicinity factor is preferred. This models the biological tendency of fish to exploit favorable currents during flight, reducing energy expenditure and increased escape speed. The influence of the current will be the strongest during selfish escape, and weakest during normal schooling. This is due to the fact that the additional speed the current provides to the fish could be crucial for an escape, but not necessary for schooling logic.

Predator Cell Atomic Model

The formal specification for the atomic Cell-DEVS model is defined as follows:

$$< X, Y, S, N, d, \tau, \delta_{int}, \delta_{ext}, \lambda, D >$$

- S: As defined in *State Set*
- $X = \{x | x \in S\}$
- $Y = \{y | y \in S\}$
- $N = S \quad \rightarrow \{(dx, dy) | \, |dx| + |dy| \leq 6\} \; for \; range \; 6 \; Von \; Neumann \; Neighborhoods$
- $D = 1 \; time \; unit$
- $\tau: N \rightarrow S$

Local Computation Function

The predator's local computation function implements pursuit and target selection:

- **Target selection:** The predator scans its range-6 neighborhood for fish, selecting the nearest fish as its target. When multiple fish are at equal distance, the predator selects randomly among them, and in modeling the confusion effect denser groups of fish force more frequent target switching. Additionally, each additional fish within the predator's vision range linearly contributes to the possibility of overwhelming the predator, causing it to stall for a time step. This is discussed further below.
- **Pursuit:** The predator moves one cell per timestep toward its target. Its turning ability is limited: it can change direction by at most 90° per timestep (it cannot reverse direction in a single step), reflecting the Zheng et al.'s constraint that predators have inferior turning ability compared to prey [2].
- **Capture:** When the predator occupies the same cell as a fish (or an adjacent cell containing a fish with no escape route), the fish is captured and removed from the simulation.
- **Current resistance:** The predator's movement is partially influenced by water currents but at reduced effect compared to fish. When moving against a strong current (vicinity factor ≤ -0.5), the predator may be slowed (skip a movement step), simulating the greater drag on its larger body.

Asymmetric Cell-DEVS Coupled Model

The coupled model defines a 20x20 cell space with asymmetric neighborhoods and vicinity factors. The scenario is configured through a JSON file with the following structure:

Cell Space and Topology

Unlike the symmetric model, cells are identified by string labels (e.g., (0,0) through (19,19)). Each cell's neighbor list is explicitly defined, enabling the predator cell to have a different (larger) neighborhood than fish cells while sharing the same coupled model. The grid is not wrapped — boundary cells have fewer neighbors.

Current Zone Definition

The cell space is partitioned into current zones. In the default configuration, an eastward current occupies the upper third of the grid (rows 0–6), a calm zone occupies the middle third (rows 7–13), and a westward current occupies the lower third (rows 14–19). The vicinity factors on horizontal connections within current zones are set to ±1.0 (assisted/opposed), while vertical connections and connections in the calm zone are set to 0.0.

Initial Conditions

Fish are initially distributed across the grid in configurations specific to each test scenario. The predator is initialized at a fixed position outside the initial fish school and assigned an initial movement direction toward the school. Multiple scenarios are provided to test different aspects

of the model: schooling without predator, predator approach from various directions, predator attack with and without current effects, and large-school evasion dynamics.

Summary of Model Parameters

Parameter	Value	Justification
Grid size	20 × 20	Sufficient space for evasion maneuvers
Fish neighborhood range	4	Range 2 schooling + range 4 predator detection
Predator neighborhood range	6	Superior predator perception [2]
Cooperative escape trigger	Range 3	Mapped from $d_{care} \approx 0.6 \times d_{vis}$ [2]
Selfish escape trigger	Range 1–2	Mapped from d_{urg} [2]
Patience zone	Range 4	$\varepsilon_{pat} = 0.4$ [2]
School cohesion weight	0.7	Optimal value from Zheng et al. [2]
Swerve behaviour	Perpendicular	Mapped from $\theta_{urg} \approx 30°$ [2]
Current vicinity factor	±1.0	Strong current; ±0.5 for moderate
Delay function	1 time unit	Transport delay, synchronized updates

Table 1: Summary of Model Parameters

Simulation Results

This section will present the simulation results of four distinct testing phases. Phase A validates the structural integrity of the simulation. Phase B simulates scenario-specific behavioural assertions for each of the behaviour patterns. Phase C runs various simulations to evaluate predator capture dynamics under stochastic conditions. Phase D isolates current dynamics, such that we are able to view the impact of current presence and absence in similar situations.

To implement this testing framework, a Python script was written to automate testing. The decision to automate testing was made due to the complexity of the grid space, as well as the random nature of some of the components. With this framework, we are able to run the test suite multiple times to verify that the top model is in fact performing as intended and without error.

For the sake of keeping a reduced page count, some simulation/scenarios will just be described and their results discussed, while others will be visualized. Full log and configuration files can be found in the attached GitHub repository, as well as videos of each tested scenario.

For clarity, the following screenshots were taken from running the scenarios in the DEVS Simulation Viewer [6]. The state variables appear in the following order: Top Grid: {Presence, Direction, Orientation}, Bottom Grid: {Behaviour, predatorDist, predatorDir}. The following color scheme was used:

$$Presence: 0 \rightarrow grey, 5 \rightarrow red, 10 \rightarrow black$$

$$Direction: 0 \rightarrow grey, 1 \rightarrow green, 2 \rightarrow yellow, 3 \rightarrow blue, 4 \rightarrow pink$$

$$Orientation: 0 \rightarrow grey, 1 \rightarrow green, 2 \rightarrow yellow, 3 \rightarrow blue, 4 \rightarrow pink$$

$$Behavior: 0 \rightarrow grey, 1 \rightarrow orange, 2 \rightarrow purple$$

$$predatorDist: 0 \rightarrow grey, 1 \rightarrow red, 2 \rightarrow light\ red, 3 \rightarrow orange, \rightarrow yellow, 5 \rightarrow light\ green, 6 \rightarrow green$$

$$predatorDir: 0 \rightarrow grey, 1 \rightarrow green, 2 \rightarrow yellow, 3 \rightarrow blue, 4 \rightarrow pink$$

Framework Setup

Each scenario used in testing was generated using a Python script that produces an asymmetric Cell-DEVS JSON file for a 20x20 grid space. The script parameterizes initial fish placement, predator placement, and zone current configurations via vicinity factors. Every scenario is generated with two variants, one with currents and one without. For those that do have currents, the zone split is as described above. By having these variants, we are able to directly compare the impact of currents versus no currents under the same scenario.

The simulation will read the JSON configuration, run for a specified time, and output to a CSV log. The test suite will examine these logs and perform assertions against them to verify the performance of the simulation in the given scenario.

It is important to note that there are three sources of stochastic behaviour within the model. This means that certain outcomes become probabilistic rather than deterministic. These sources will be described in the following subsections as they influence the experimental results.

Probabilistic Mechanisms

Cooperative Escape Reluctance

During cooperative escape, each fish independently computes a probability at each timestep that determines if they are to flee in the consensus direction or hold in place. The probabilities for each are set at 0.8 and 0.2, respectively. This mechanic serves two primary purposes. First, it reduces the school's effective flee speed in comparison to the predator's speed of pursuit. This ensures that a predator is able to close the gap between itself and the school over time, and eventually capture a fish. Without this mechanic, a fleeing school moves at the same speed as the predator, leading to the predator never being able to close the distance gap in open water, and rather having to chase the school to the boundary of the cell space. Secondly, because the computation is independent per fish, there is a chance that some fish will stall and others will flee, causing the school to fracture. Due to the school fragmentation, we introduce "stragglers" that can become isolated targets. This also adds to the biological realism of the system, as fish at the opposite end of a school may not feel as threatened by the predator and will stay as a more cohesive unit, while those closer to the predator feel more threatened and are more susceptible to fracturing.

In the *Challenges* section of this report, the decision to introduce this mechanic will be discussed in more depth.

Selfish Escape Panic

When a fish's behaviour enters selfish escape, there is a 30% chance that they will panic and anchor in place instead of fleeing off the predator's line of attack. This allows the predator to

close the distance gap between itself and the prey. This mechanism is inspired by a predators lunge dynamic as modeled in Zheng et. al, where the prey's reaction time is finite [2]. Having a 70% swerve rate still allows the fish to evade the predator when isolated from its school, but it also leads to fish eventually getting captured as they anchor in panic.

Predator Confusion Stall

A predator's target selection is influenced by the number of fish in its visible neighbourhood. The probability of a confusion-induced stall scales linearly at 6% per visible fish, and is capped at 36% to ensure the predator does not anchor in place for an extended period. When schools are on the larger size, six or greater, we see a predator stall nearly a third of the time. When schools are smaller, or fish are fragmented, we see a more effective pursuit of the prey. This mechanism is taken from Zheng et al, where predator capture success falls as the local fish density increases [2].

Phase A: Structural Testing

This phase highlights several invariants over all the identified scenarios. These invariants hold for every timestep of every simulation ran, and confirm that the model's state transitions are consistent regardless of behavioural mode, current cell population density, vicinity factors etc.

Invariant	**Description**
Valid States	Each of the state fields is valid at every timestep.
Empty Cell Consistency	Empty cells have all fields cleared to zero.
Boundary Respect	No entity occupies a cell outside the 20x20 cell space.
Predator Conservation	The predator remains constant throughout the simulation (does not disappear).
Fish Conservation	Fish counts never increase, they only decrease on capture.
Capture Tracing	When fish count drops, a predator is also arriving at a cell that was previously occupied by a fish.
BP Correctness	Schooling fish never have active predator alarms, alarmed fish are never in schooling mode. This corresponds to behaviour != 0 if and only if predatorDist > 0.
Predator Fields	predatorDist and predatorDir are paired, and predatorDist > 0 if and only if predatorDir != 0
Movement Consistency	When an entity leaves a cell, it appears in the adjacent cell as indicated by its prior direction.

Table 2: Structural Testing Breakdown

Running the test suite reveals that all invariants hold across all simulations. The most important note of this phase is that capture traceability indicates that the capture mechanic is working as intended; that being that no fish are unexplainably leaving the simulation and instead are being captured and removed accordingly.

Phase B: Scenario Specific Assertions

In this phase, each scenario is tested against targeted assertions. Each scenario is designed to verify a specific aspect of the model.

Schooling Without a Predator

The schooling scenarios initialize a set of fish across the grid, without a predator present. By removing the predator from the scenario, we are able to verify that the changes made to the original Cadmium implementation are not affected by the added evasion logic. The initial configuration is as below.

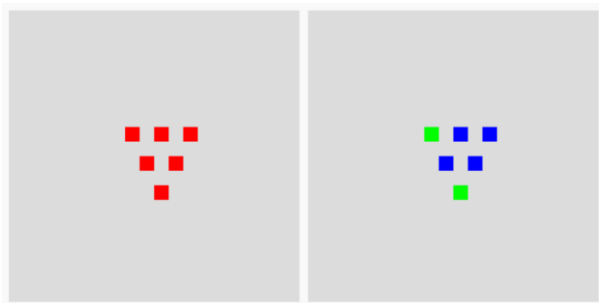

Fig. 2. Schooling Scenario at $\boldsymbol{t = 0}$

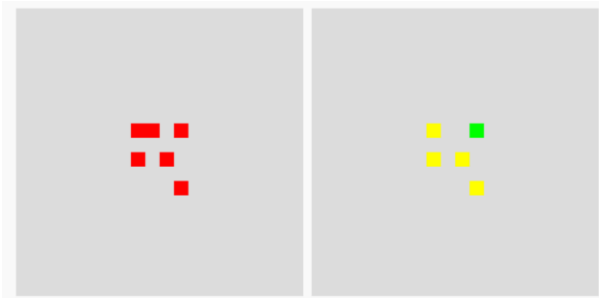

Fig. 3. Schooling Scenario at $\boldsymbol{t = 2}$

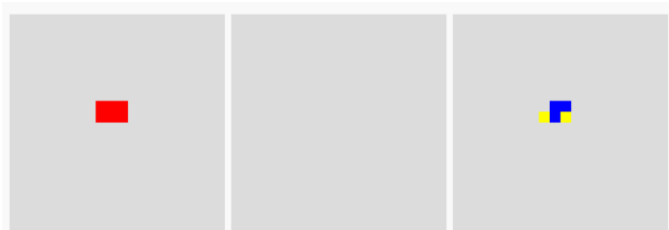

Fig. 4. Schooling Scenario at $\boldsymbol{t = 12}$

As seen from the above sequence, the schooling logic still performs as intended.

Predator Approach From All Directions

To cover a predator attacking from all angles, we devised four scenarios; *predator_east*, *predator_west, predator_south*, and *predator_north*. In each of these scenarios, a school is placed in the center of the cell space, and a predator is initially placed according to its direction of attack (e.g. *predator_north* places the predator initially in the northern area of the cell space.).

This test confirms that cooperative escape behaviour is observed at least once as the predator enters the cautionary zone, and that at least one capture occurs. These scenarios demonstrate the distance-based behaviour switching rule, and how the threshold performs according to the rule

set. This sees fish continue schooling when the predator is at range 4, cooperative escape triggers at range 3-4, and selfish escape triggers at range 1-2.

An example scenario is seen below, showcasing the *predator_east* scenario.

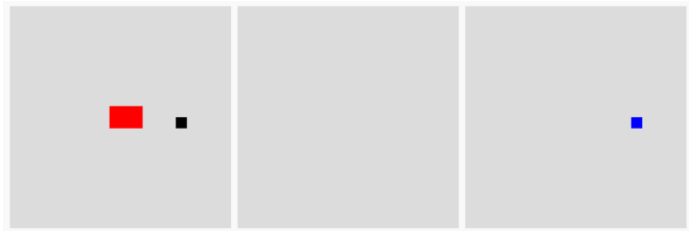

Fig. 5. Predator East Scenario at $\boldsymbol{t = 0}$

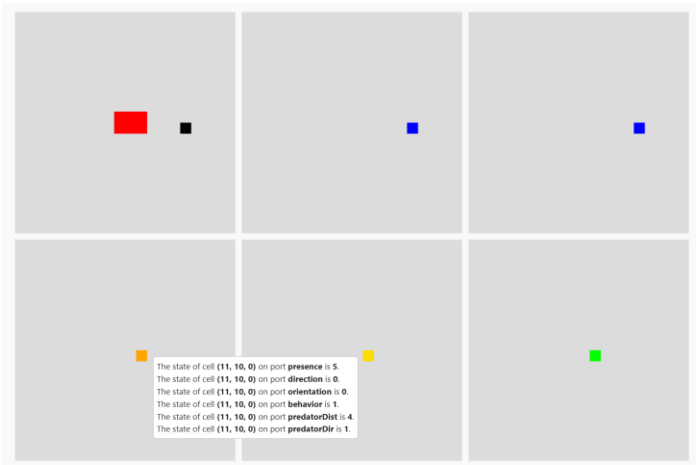

Fig. 6. Predator East Scenario at $\boldsymbol{t = 1}$

Fig. 6 demonstrates how a singular fish has detected the predator and wishes to begin cooperative escape.

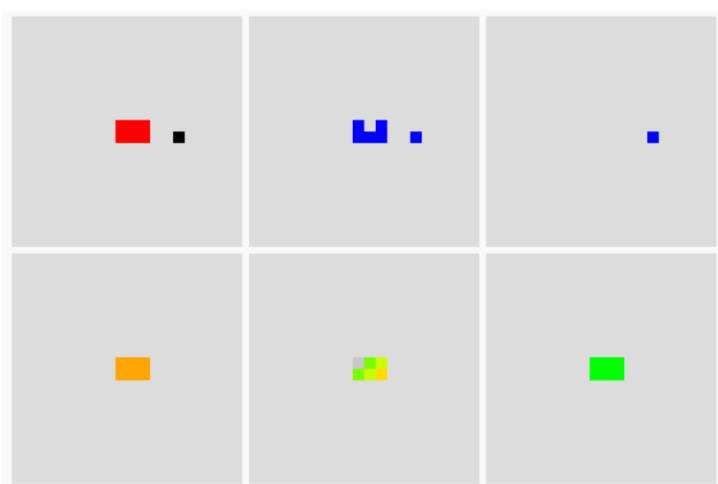

Fig. 7. Predator East Scenario at $\boldsymbol{t = 2}$

In Fig. 7 we now see that the singular fish in the previous time step has propagated the presence of a predator to the rest of the school, who have now also entered cooperative escape behaviour.

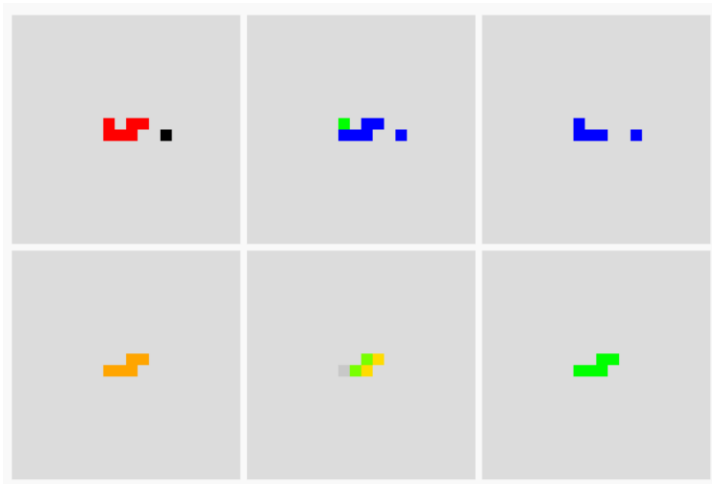

Fig. 8. Predator East Scenario at $\boldsymbol{t = 3}$

In Fig. 8 we see the first school fracture.

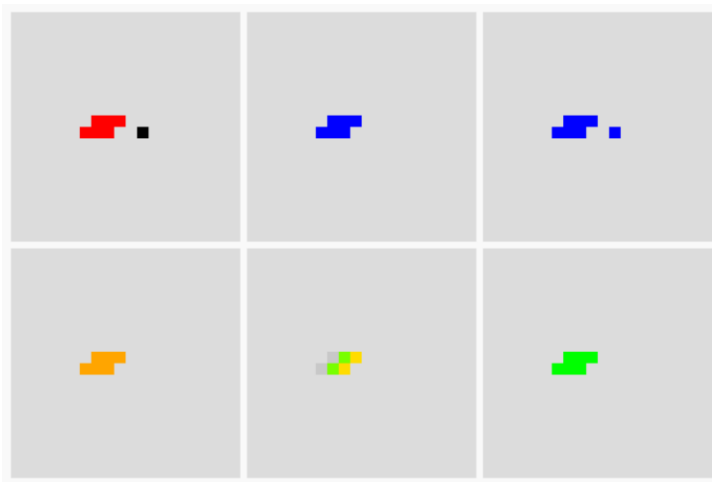

Fig. 9. Predator East Scenario at $\boldsymbol{t = 5}$

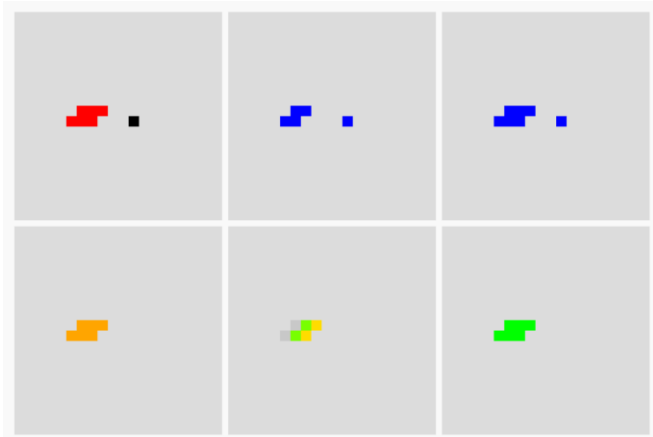

Fig. 10. Predator East Scenario at $t = 6$

Between t=5 and t=6, we see that the predator was confused and anchored in place. It will now resume its pursuit.

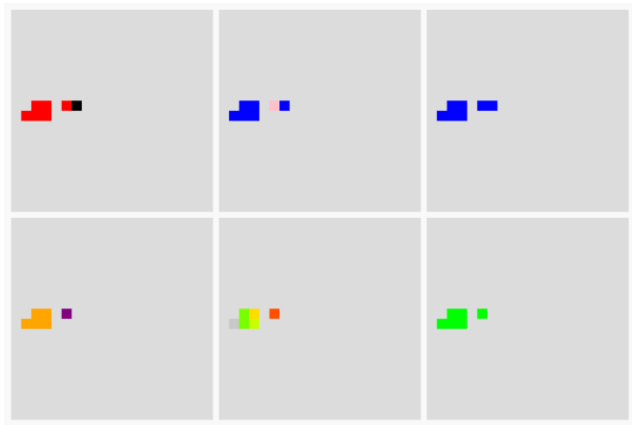

Fig. 11. Predator East Scenario at $t = 12$

Here in Fig. 11, we see the first example of a fish entering selfish escape as a predator is directly to the East.

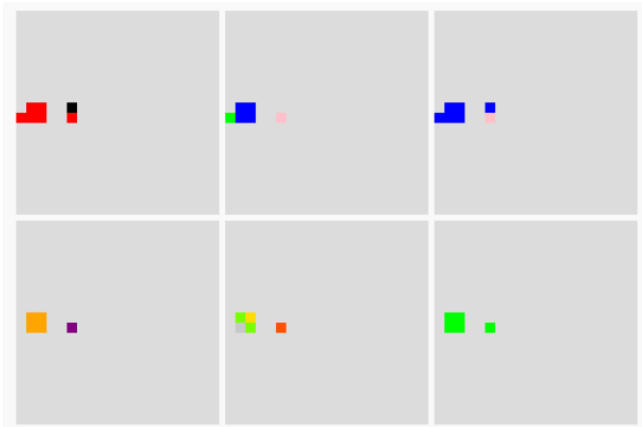

Fig. 12. Predator East Scenario at $t = 13$

In Fig. 12 we see that the fish exhibiting selfish escape behaviour has swerved south, off the predators line of attack.

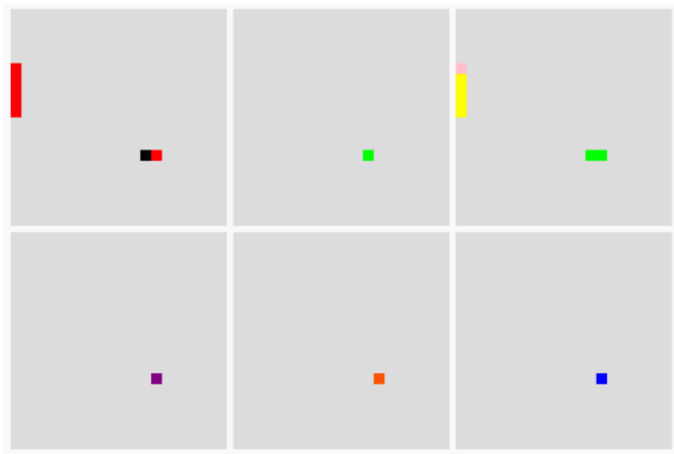

Fig. 13. Predator East Scenario at $t = 33$

For the sake of this report, we will jump to t=33, where we see that the predator has chased a target fish into the South-East portion of the cell space. Additionally, we see that a new school has formed with the remaining five fish along the western border.

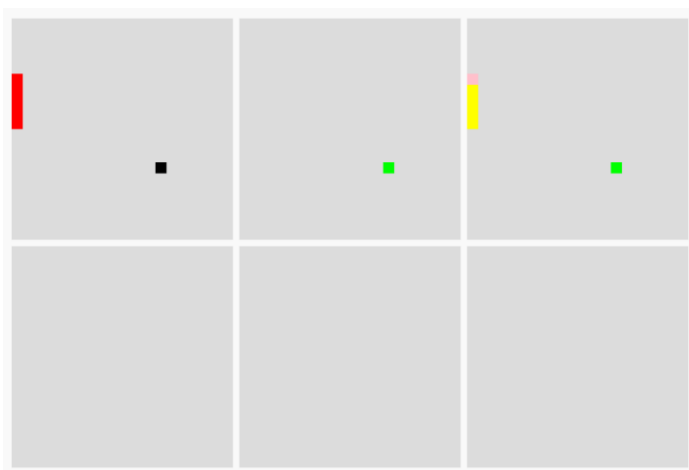

Fig. 14. Predator East Scenario at $t = 34$

In Fig. 14 we see that the fish being pursued by the predator panicked while exhibiting selfish escape behaviour, and as such was captured by the predator.

Cooperative Escape

Three cooperative escape scenarios were tested to examine school movement under certain types of predator pressure. The *coop_platoon* scenario arranges fish in a horizontal line, testing vacating movement where a chain of fish all move in the same direction. The *coop_diagonal* scenario places fish in a diagonal line relative to the predator, testing that the consensus flee direction correctly propagates through the school. Finally, the *coop_wall* scenario places the school against the boundary, testing fallback movement logic that sees the fish choose a

perpendicular direction instead. Each of these scenarios confirm that cooperative escape is triggering for at least one of the fish.

The logs and videos have been included in the repository.

School Reluctance

This scenario is used to demonstrate the school's reluctance to move despite an approaching predator. The test checks that at least one fish that is exhibiting cooperative escape behaviour chooses to anchor in place. Due to this mechanism being stochastic, it is possible to run the simulation and not see the desirable outcome. As such, in Phase C we run the simulation 20 times to observe the average behaviour of the school. The result of this testing confirms that schools are indeed reluctant to move appropriately 20% of the time.

Phase C: Statistical Testing

This phase was designed to address the stochastic outcomes that cannot be verified in a single test run. To address this, each test was run 20 times to evaluate the average metrics.

Capture Probability

To observe the probability of at least one fish being captured during a simulation, the reluctance scenario was run 20 times over 40 time steps, setting a threshold of having at least one capture in 10 of the 20 runs, or at least 50% of the time. This was met, with a run producing 117 captures with at least one per simulation.

Additionally, the *large_school* scenario was run 20 times over 60 time steps, with the goal of observing a capture in at least 80% of the runs. This threshold was also met, with a given run producing 202 captures, with at least one per simulation.

Between these two scenarios, we can confirm that the reluctance mechanism does in fact produce captures at a steady rate. The absence of such mechanism will be discussed in the challenges section.

Open-Water Capture Rate

A main driving factor of the introduction of stochastic mechanisms was to encourage open-water captures – which can be defined as having a Manhattan distance of at least three from the border. To test the effectiveness of the stochastic mechanisms, we set a threshold of 30% of all captures occurring in open-water across all statistical runs. The same scenarios used in *Capture Probability* were used here, and did in fact meet the threshold, with a given run producing a 39.3%, and 51% open-water capture rate, respectively.

This allows us to conclude that we have met the design goal motivated by Zheng et al., that being that predators can succeed through the confusion effect to capture fish in open-water and not just corner them against the border.

Phase D: Current Testing

Cooperative Escape – Current Influence

The *current_coop_fallback* scenario places three fish in the first row on the grid space, which features an eastward current zone. Additionally, there is a predator placed South of the fish. The point of this test is to observe that schools choose to travel in current-aided directions when fleeing a predator. To observe this, we will show some sample time steps.

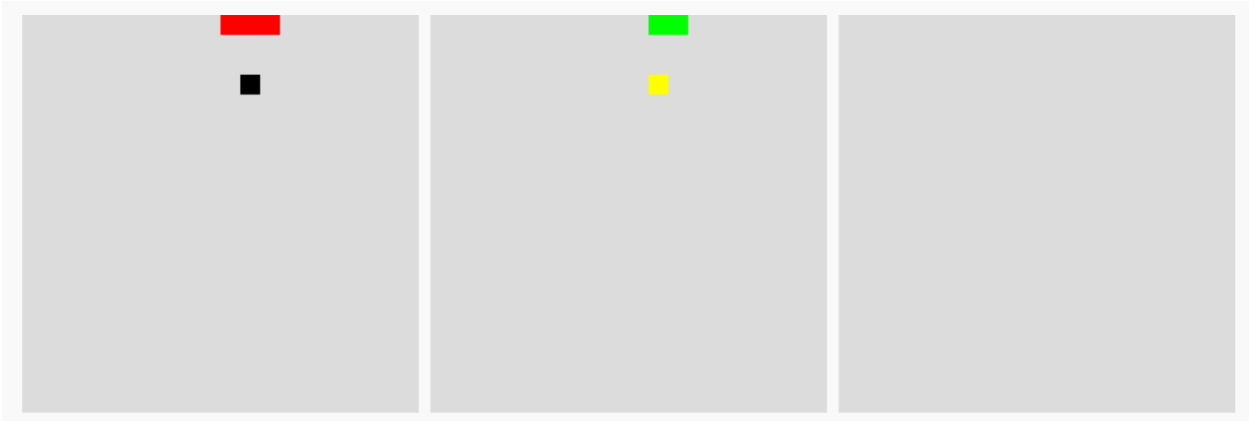

Fig. 15. Current Influence Scenario at $t = 0$

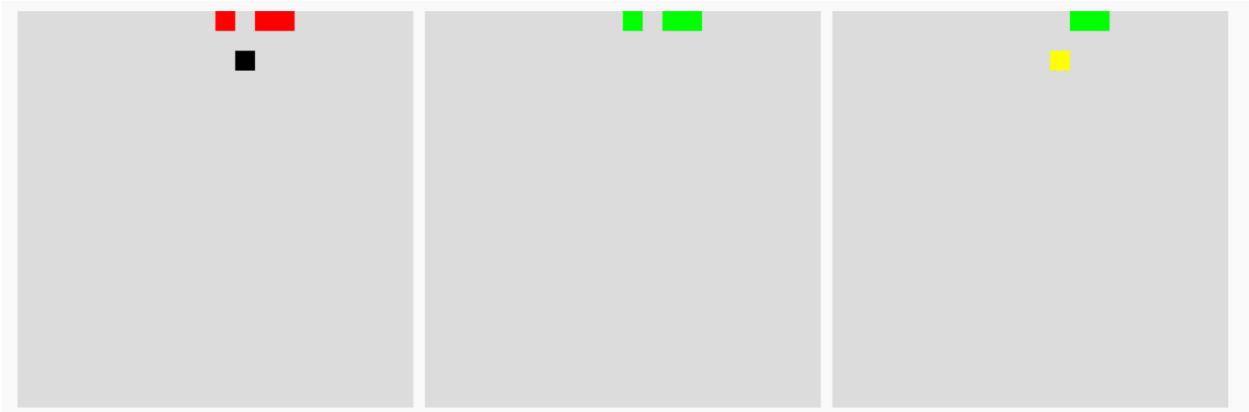

Fig. 16. Current Influence Scenario at $t = 1$

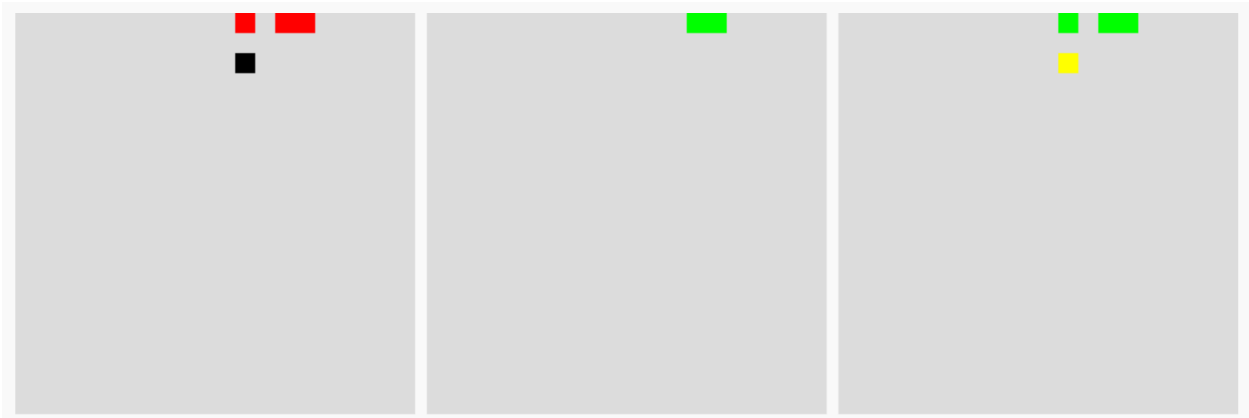

Fig. 17. Current Influence Scenario at $t = 2$

Looking at these three time steps, we confirm the correct behaviour. In comparison, we can examine the exact same scenario but without current. This results in the following.

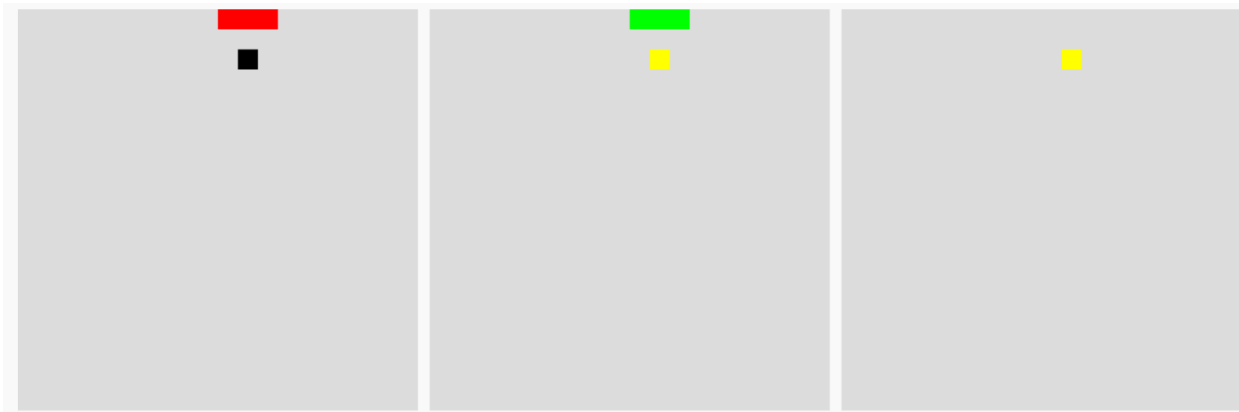

Fig. 18. No Current Influence Scenario at $t = 1$

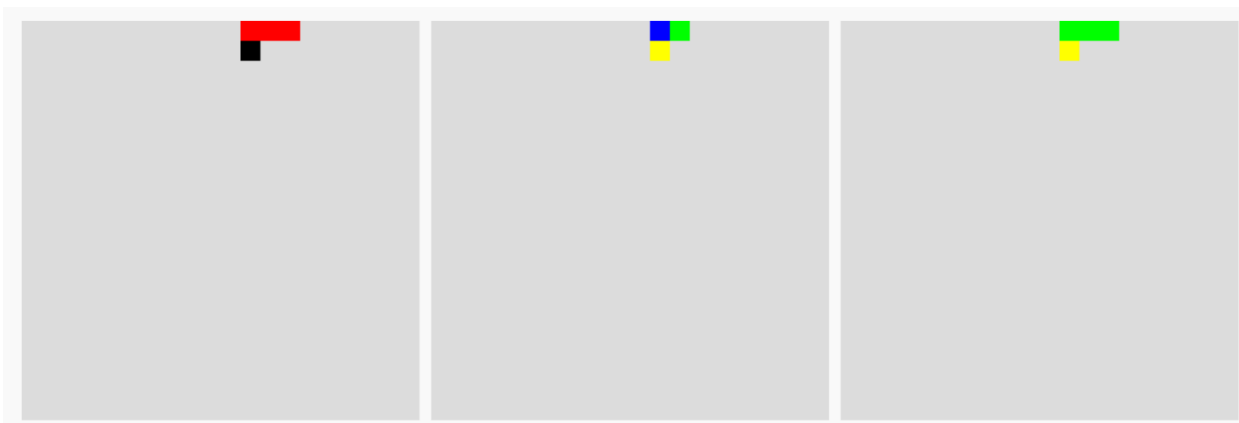

Fig. 19. No Current Influence Scenario at $t = 2$

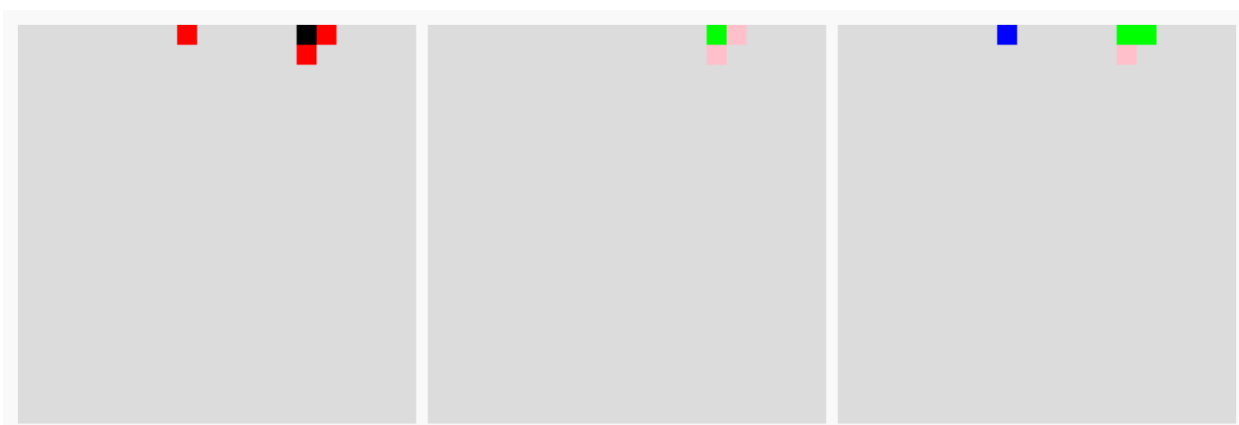

Fig. 20. No Current Influence Scenario at $t = 6$

Through these time steps we see the school initially moving eastwards, before branch West and South. Allowing this simulation to continue to run we observe that both the predator and fish

cells move with absolute freedom with the absence of the current, as they travel in all directions at random when within the top rows of the grid space. This demonstrates the impact of currents on fish movement, and lack of influence in their absence.

Large Fish School Scenario

Perhaps the best scenario to see how each mechanism interacts is the *large_school* configuration. In this configuration, we see an initial school that is a 6x6 grid of fish with a singular predator to the East. This configuration reveals how predator detection propagates through the school, as with such a large school not each fish is made aware of the predator, and rather only a subset is. This is a great example to showcase the limits of alarm propagation.

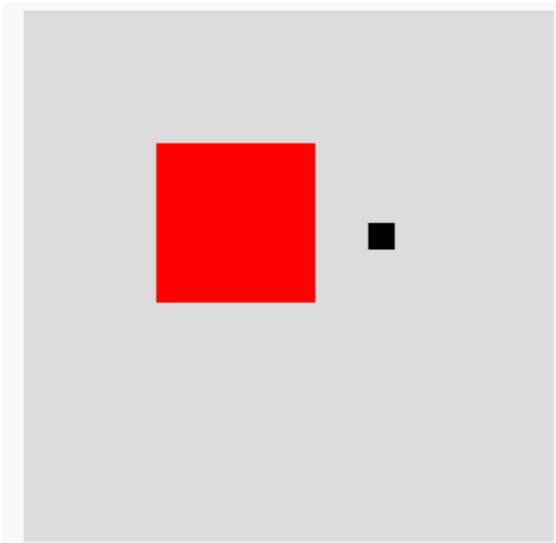

Fig. 21. Large School Scenario at $t = 0$

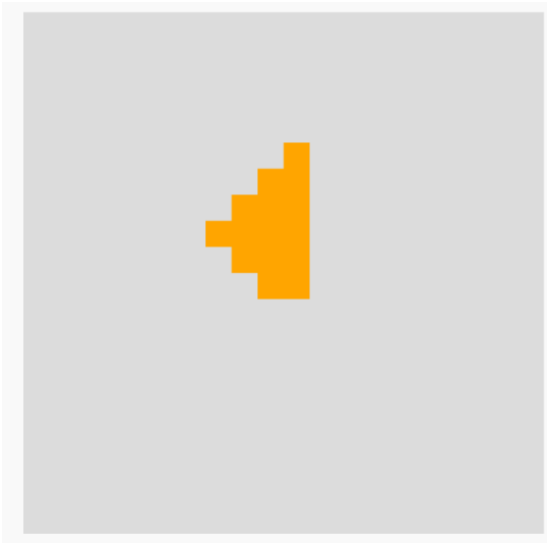

Fig. 22. Large School Scenario at $t = 0$ ***(Alarm Relay)***

As this simulation progresses, we see all the main components working together; the predator becoming confused by the large school, fish anchoring while in cooperative escapes rather than fleeing, fish entering selfish escape and more.

Fig. 23. Large School Scenario at $t = 0$***, 9, 19, 65, and 99***

From the series of time stamps in Fig. 23 we can see how the predator is able to fracture the initial school and isolate target fish. Additionally, we can see how new schools form following the initial fragmentation, and continued predator pursuit.

All in all, the result is a highly fractured fish school as the predator charges head on into the school. The full log, configuration, and visualization can be found in the repository.

Summary of Test Results

Overall, the test suite demonstrates that the model is performing as intended by verifying each component of the model, including:

- Unchanged schooling logic
- Distance-based behaviour rules
- Probabilistic reluctance and panic
- Predator awareness relay
- Current-influenced movement patterns
- Predator confusion and current resistance

Challenges

There were various limitations and challenges that became apparent throughout the development of this model. The biggest will be highlighted in this section.

Behaviour in Open-Water

The biggest challenge faced in the development of this model was that often simulations fell into a pattern of the predator chasing the fish around the grid space until they were caught along a border or corner. This was occurring due to the predator and fish cells being modeled with the same movement speed – one cell per time step. To overcome this challenge, we introduced the probabilistic mechanisms of reluctance and confusion. Adding these mechanisms into the model slowed the movement of both fish and predators, which lead to more interactions in the body of the grid space as opposed to along the border. As a result, this greatly increased the realism of the model.

Conclusion

This work presented an asymmetric Cell-DEVS model of fish school formation and predator evasion, implemented in the Cadmium simulation framework. Building on Etemad's initial symmetric formulation, the work introduced three primary extensions: conversion to asymmetric Cell-DEVS, the addition of a predator and distance-based behaviour patterns, and environmental water currents as directional vicinity factors.

The distance-based adaptation of Zheng et al. produced the intended three-pattern evasion dynamics, with alarm relay ensuring predator awareness propagated through the school beyond the direct detection range. The introduction of probabilistic reluctance, panic, and predator confusion mechanisms was necessary to produce ecologically plausible dynamics, as without them, equal movement speeds caused simulations to resolve almost entirely at grid boundaries rather than open water. Statistical testing confirmed that at least 30% of captures occurred in

open water, validating the design goal. Current vicinity factors further demonstrated a meaningful influence on movement, with fish consistently favoring current-assisted escape directions and predators exhibiting reduced mobility against strong opposing currents.

Overall, the model demonstrates that with a small set of local rules and distance-based thresholds, we can sufficiently reproduce collective evasion dynamics as described by Zheng et al. within the Cell-DEVS formalism.

References

[1] M. Etemad, “Modeling the Formation of School of Fish.” Carleton University, Nov. 2014.

[2] M. Zheng, Y. Kashimori, O. Hoshino, K. Fujita, and T. Kambara, “Behavior pattern (innate action) of individuals in fish schools generating efficient collective evasion from predation,” *J. Theor. Biol.*, vol. 235, no. 2, pp. 153–167, Jul. 2005, doi: 10.1016/j.jtbi.2004.12.025.

[3] S. Hubbard, P. Babak, S. Th. Sigurdsson, and K. G. Magnússon, “A model of the formation of fish schools and migrations of fish,” *Ecol. Model.*, vol. 174, no. 4, pp. 359–374, Jun. 2004, doi: 10.1016/j.ecolmodel.2003.06.006.

[4] I. Aoki, “A simulation study on the schooling mechanism in fish.,” *NIPPON SUISAN GAKKAISHI*, vol. 48, no. 8, pp. 1081–1088, 1982, doi: 10.2331/suisan.48.1081.

[5] A. Huth and C. Wissel, “The simulation of the movement of fish schools,” *J. Theor. Biol.*, vol. 156, no. 3, pp. 365–385, Jun. 1992, doi: 10.1016/S0022-5193(05)80681-2.

[6] “DEVS Simulation Viewer | ARS Lab | Carleton University.” Accessed: Apr. 23, 2026. [Online]. Available: https://devssim.carleton.ca/cell-devs-viewer/